VEDATECHNOLOGY VIRTUAL INTERNSHIP 2026

TASK 16: HTTP AND HTTPS

Track: Cyber Security
Task Title: HTTP and HTTPS
Student Name: Nidhi Patil
Internship Organization: Vedatechnology
Objective: To compare HTTP and HTTPS and understand the security benefits of TLS.

1. Introduction

HTTP stands for Hypertext Transfer Protocol. It is used to transfer data between a web browser and a web server.

HTTPS stands for Hypertext Transfer Protocol Secure. It uses TLS (Transport Layer Security) to protect communication between the browser and the server.

2. Objective

- To understand HTTP and HTTPS.
- To inspect web requests using Chrome Developer Tools.
- To compare their security features.
- To understand the benefits of TLS.

3. Tools Used

1. Chrome Developer Tools
2. Google Docs

4. Practical Implementation

Practical 1: HTTP Request Inspection

Website: NeverSSL
URL entered: http://neverssl.com

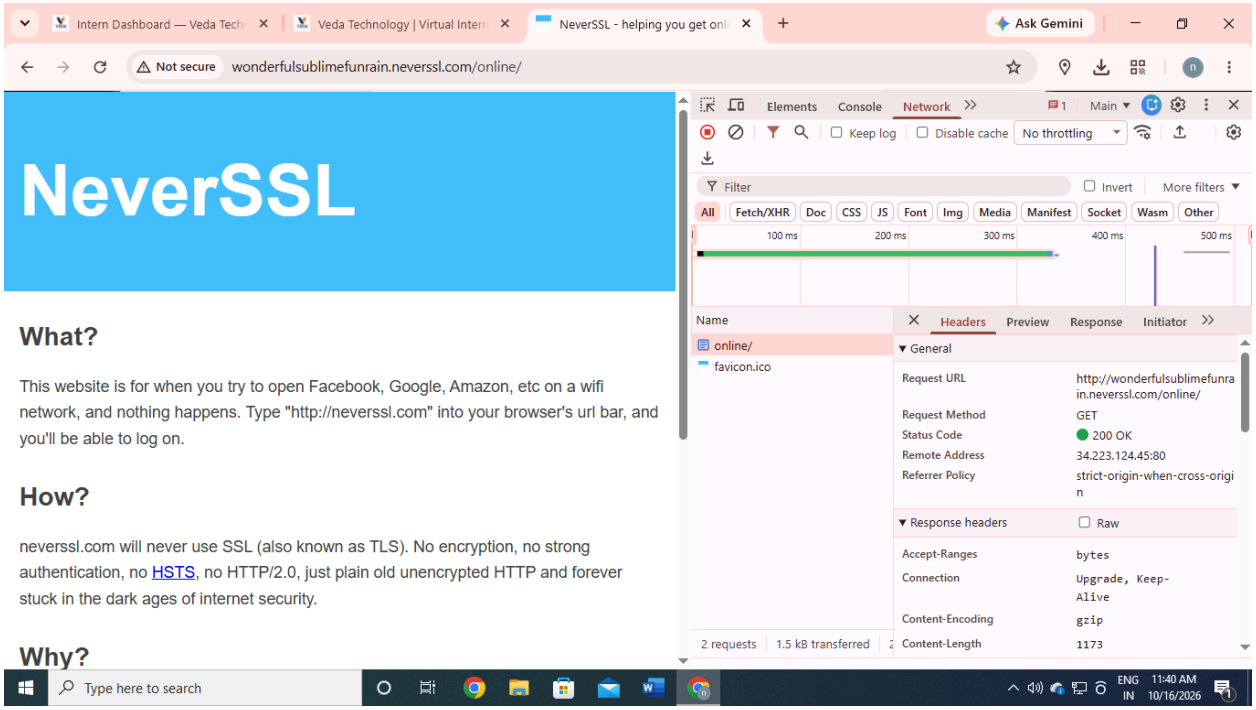
Procedure:

1. Opened the NeverSSL website in Google Chrome.
2. Opened Developer Tools by pressing F12.
3. Selected the Network tab.
4. Reloaded the webpage.
5. Selected a network request and opened Headers.
6. Inspected the Request URL, Request Method and Status Code.

Observation:

The selected request used HTTP. The Request Method was GET, the Status Code was 200 OK, and the browser displayed a “Not secure” warning.

Screenshot 1: HTTP Practical



Practical 2: HTTPS Request Inspection

Website: Example Domain
URL entered: https://vedatechnology.in/

Procedure:

1. Opened the Example Domain website in Google Chrome.
2. Opened Developer Tools using F12.
3. Selected the Network tab.

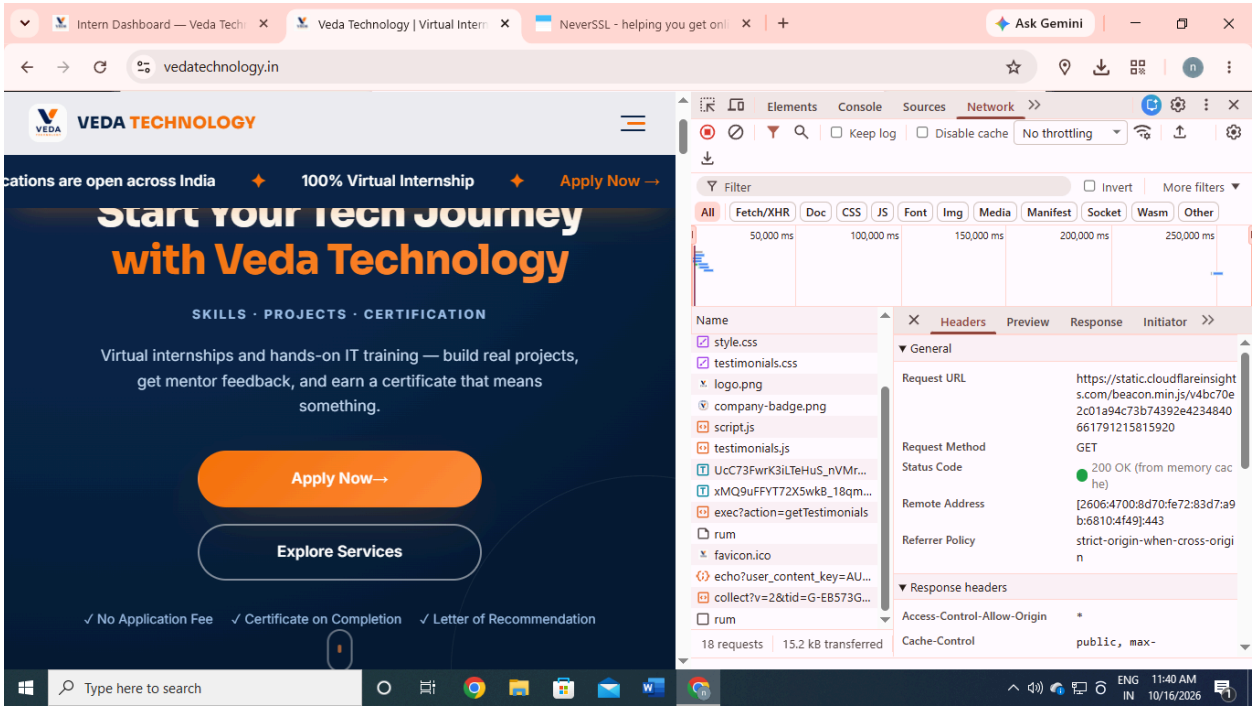
4. Reloaded the webpage.
5. Selected a network request and opened Headers.
6. Checked the Request URL and Status Code.

Observation:

The Request URL began with HTTPS. HTTPS uses TLS to encrypt data in transit, authenticate the server through its certificate and protect data integrity.

Status Code Observed: [Enter the actual status code from your screenshot.]

Screenshot 2: HTTPS Practical



5. Difference Between HTTP and HTTPS

Basis	HTTP	HTTPS
Full form	Hypertext Transfer Protocol	Hypertext Transfer Protocol Secure
Encryption	No TLS encryption	Uses TLS encryption
Default port	80	443
Authentication	No TLS certificate authentication	Uses a server certificate
Data integrity	No TLS-based integrity protection	TLS protects data integrity

Security	Not suitable for sensitive data	Preferred for sensitive communication

6. Security Benefits of TLS

1. Encryption: TLS makes transmitted data unreadable to people who intercept it.

2. Authentication: A valid server certificate helps the browser verify the identity of the website.

3. Data Integrity: TLS helps detect unauthorized modification of data during transmission.

4. Privacy: TLS helps protect passwords, personal information and payment details while they travel across the network.

7. Why Is HTTP Insecure for Sensitive Data?

HTTP does not provide TLS encryption. An attacker who can observe or interfere with network traffic may read or modify information transmitted over HTTP. Therefore, sensitive information should be sent through properly configured HTTPS websites.

8. Result

I inspected HTTP and HTTPS requests using Google Chrome Developer Tools. I observed the Request URL, Request Method and Status Code. The HTTP request on NeverSSL returned 200 OK and the browser displayed a “Not secure” warning. The HTTPS request was identified by its HTTPS URL.

9. Conclusion

This practical helped me understand the difference between HTTP and HTTPS. HTTP does not provide TLS protection, whereas HTTPS uses TLS for encryption, authentication and data integrity. Therefore, HTTPS is the preferred protocol for secure web communication.

Interview Questions

Q1. What is HTTPS?
HTTPS is the secure version of HTTP that uses TLS to protect communication.

Q2. What is TLS?
TLS is a security protocol that provides encryption, authentication and data integrity.

Q3. Why is HTTP insecure for sensitive data?
HTTP does not provide TLS protection, so information may be exposed or modified in transit.